

HUYNH HO

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EDUCATION

Gettysburg College

B.S. Computer Science & Mathematics, Minor in Data Science, GPA: 3.71

Gettysburg, PA

Expected May 2027

Relevant Coursework: Operating Systems, Data Structures & Algorithms, Database Systems, AI, Probability & Statistics

EXPERIENCE

Open Source Contributor, Backend Infrastructure

Apr 2026 – Present

Kerno (GirlScript Summer of Code), Kubernetes observability tool in Go

1 merged PR

- Replaced kubelet HTTP polling with a Kubernetes SharedIndexInformer scoped to the local node, fixing pod tracking that silently stopped on GKE/EKS clusters. Two in-memory indexes cut pod lookups from $O(n)$ to $O(1)$.
- Added exponential backoff (1s to 2 min) for API server failures and kept stale entries so pod enrichment keeps working in a degraded mode.

Open Source Contributor, Backend & Observability

Jun 2026 – Present

Tether (FastCrest), edge-to-cloud AI deployment CLI, 83+ stars

3 merged PRs

- Built request-ID middleware that stamps every API response with a UUID, so one request can be traced across a fleet. Merged after code review, with 5 unit tests.
- Fixed silent data loss on unclean shutdown by flushing pending writes at exit, with 6 tests.
- Restored a Prometheus metric that never emitted in production and fixed a double-counting bug.

Teaching Assistant, Calculus I

Aug 2024 – Dec 2024

Gettysburg College

Gettysburg, PA

- Led weekly problem-solving sessions for about 30 students and told the professor which topics they struggled with.

PROJECTS

YouLecture: AI Study Environment | Python, FastAPI, Claude API, Next.js, AWS [Live demo](#) | 2026

- Built a 5-agent RAG pipeline that turns a YouTube URL into an outline, summaries, and flashcards.
- Deployed on AWS Lightsail with Docker and nginx. Parallelized slow stages and cached transcripts.
- Fixed YouTube HTTP 429 rate limiting on AWS with a rotating residential proxy.

Cancer to Hell: Clinical Research Tool | Go, Anthropic Claude, PubMed, Next.js, SSE [Live demo](#) | 2026

- Built three LLM agents (evidence, guideline, risk) running concurrently as Go goroutines and streaming over SSE. A guardrail checks each citation against live PubMed.
- Deployed on AWS Lightsail with Docker Compose.

Line-Following Robot: PicarX | Python, Grayscale Sensors, State Machine

2025

- Led a student team building a line-following robot. Tuned sensor thresholds for different lighting and added recovery logic to find the line again.

TECHNICAL SKILLS

Languages: Go, Python, Java, JavaScript/TypeScript

Infrastructure: Kubernetes, Docker, AWS (EC2, Lightsail), Linux, nginx, Git

Backend & Observability: REST API design, FastAPI, Prometheus metrics, MySQL

AI: Claude API, RAG, multi-agent pipelines, Claude Code